05-13 第2篇

阅读理解（说明文） Why Does a Rainbow Have Seven Colours?

① After a summer rain, if the sun comes out quickly, you may be lucky enough to see a **rainbow** stretching across the sky. A rainbow is one of the oldest puzzles in nature. Why does it appear only after rain? Why is it always arc-shaped? And why does it show seven colours, always in the same order? The answer lies inside each small drop of rain.

② To understand a rainbow, we first need to understand **white sunlight**. Although sunlight looks plain and "white", it is actually made of many colours mixed together. We see this truth in science class when a teacher passes white light through a glass **prism**. The light comes out the other side as a bright band of colours — red, orange, yellow, green, blue, indigo and violet.

③ Every rain drop in the sky works like a tiny natural prism. When a ray of sunlight enters a drop, it slows down a little because water is denser than air. Different colours slow down by slightly different amounts. This makes each colour **bend**, or change direction, at a slightly different angle. The light then hits the back of the drop, bounces, and comes out the front — with its colours now separated in a fan.

④ Millions of drops are doing this at the same time, in slightly different positions in the sky. Our eyes catch only certain drops for each colour. The drops that send red light to our eyes are a bit higher up; those that send violet light to our eyes are a bit lower. This explains why a rainbow always has the **same colour order**, with red on the outside and violet on the inside.

⑤ The shape of a rainbow is also part of the puzzle. A rainbow is not a true bridge from one mountain to another — it is a full **circle**. We just do not usually see the bottom half of the circle, because the ground is in the way. Pilots flying in clear air above the clouds sometimes see perfect round rainbows around the shadow of their airplane.

⑥ Rainbows also need three conditions to appear: water drops in the air, sunlight behind your back, and clear distance in front of you. When you stand in the right place, you become the **centre** of your own rainbow. A friend standing next to you sees her own, slightly different one. This is a lovely scientific idea hidden in a beautiful sky: every person has a rainbow that belongs only to them.

根据短文内容，选择最佳答案。

1. What does the passage say about sunlight?

A. Sunlight is only yellow.
B. Sunlight looks white, but it is actually made of many colours mixed together.
C. Sunlight has exactly three colours.
D. Sunlight has no colours at all.

2. What does each rain drop in the sky do to sunlight?

A. It turns sunlight into water.
B. It works like a tiny prism and bends different colours at different angles.
C. It makes sunlight dark.
D. It sends sunlight back to the sun.

3. Which colour is on the outside of a rainbow?

A. Red.
B. Green.
C. Indigo.
D. Violet.

4. What is the real shape of a rainbow?

A. A line.
B. A square.
C. A full circle; we usually see only half of it because the ground is in the way.
D. A triangle.

5. What three conditions are needed for you to see a rainbow?

A. Sunlight in your eyes, water on your feet, wind in front.
B. Rain drops in the air, sunlight behind your back, clear distance in front of you.
C. Only sunlight, and nothing else.
D. Only a camera and a phone.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "sunlight looks plain and 'white', it is actually made of many colours mixed together."
【解析】看似白光其实由多色混合而成。A/C/D 均不符。故选 B。

2.
【答案】B
【题型】细节理解题

【定位】第③段 "Every rain drop in the sky works like a tiny natural prism... makes each colour bend..."
【解析】每个水滴像小三棱镜折射光线。A/C/D 均与原文相反或无据。故选 B。

3.
【答案】A
【题型】细节理解题
【定位】第④段末句 "red on the outside and violet on the inside."
【解析】外侧是红、内侧是紫。B/C/D 均不符。故选 A。

4.
【答案】C
【题型】细节理解/推理判断题
【定位】第⑤段 "A rainbow is not a true bridge... it is a full circle."
【解析】彩虹实为完整圆圈，地面挡住下半。A/B/D 均不符。故选 C。

5.
【答案】B
【题型】细节理解题
【定位】第⑥段 "water drops in the air, sunlight behind your back, and clear distance in front of you."
【解析】三条件：空中水滴、背对太阳、前方开阔。A/C/D 不完整或错误。故选 B。

词汇注释（超纲词）

- **prism** [n.] 三棱镜
- **indigo / violet** [n.] 靛色 / 紫色
- **ray** [n.] 光线